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Chapter
12

Cultivation Technology of Tomato in Greenhouse

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Abstract

India is the fifth largest producer of tomato accounting for 6.0% of the world production. It is considered as world's largest vegetable crop after potato and sweet potato. Among all the vegetable crops, tomato is the first crop that was grown in greenhouse. It is relatively easy to grow compared to other vegetables like pepper and cucumbers and also the fruit yield is very much high under protected cultivation. The major growing states are Odisha, Andhra Pradesh and Karnataka. Tomato (*Lycopersicon esculentum*) considered as "poor man's orange" and universally treated as "Protective Foods". There are two types of tomato determinate or bush tomato and indeterminate or vine tomato. For the sake of greenhouse systems mostly indeterminate type are used for maximizing crop productivity and increasing the quality of vegetables produce. They produce flowers and fruits continuously along the main stem as they grow.

Keywords: Tomato, protected cultivation, fertigation, nursery management and harvest.

1. Introduction

Tomato belongs to Mexican and Peruvian region. It is a herbaceous plant that grows to a height of 1-3 m with a weak woody stem. It is universally treated as 'Protected Food'. It is rich in vitamin C-31 mg/100g. The flowers are yellow in colour and the fruits cultivated varieties vary in size from cherry tomatoes about 1-2 cm in size to beefsteak tomatoes, about 10cm or more in diameter. There are 700 varieties of tomatoes cultivated throughout the world. It is considered as important commercial and dietary vegetable crop, as it is a short duration crop and gives high yield, it is important from economic point of view and hence area under its cultivation is increasing day by day.

2. Reasons for Preferring Protected Cultivation of Tomato

Presently, progressive farmers are adopting commercial protected cultivation of high value vegetables and flowers (Maitra *et al.* 2020). Following are the reasons to prefer protected cultivation of tomato.

- Better quality produce.
- Higher productivity.
- Nursery raising and hardening of plants.
- Better insect and disease control and reduced use of pesticides.
- Off-season cultivation.
- Efficient use of resources.

2.1 Importance of Green House Production

Green house tomato food production offers to produce marketable product at times when supply is low. It increases the length of time tomatoes are available. Greenhouse tomatoes are sold at higher price than field tomatoes (Fig. 1). System offers profitable opportunities in the spring before field tomatoes are harvested and, in the fall, when field tomatoes depleted. (February-March)Plant set more fruit per plant and the heating cost is less. Light intensity and duration increase as outside temperature increase, making it less costly to heat and provide ventilation and humidity control. All of these result in increased yields making spring production more profitable than fall production.

Fig. 1: Protected cultivation of tomato in open ventilated polyhouse

3. Cultivation Practices in Green House

3.1 Climate

Tomato is a warm season plant. It can withstand severe frost conditions. The minimum temperature for germination of seeds ranges from 8°C to 10°C. The night temperature is the critical factor in fruit setting with the optimum range of 16°C-22°C. Fruits fail to set at or below 12°C. When the average maximum day temperature goes above 32°C and average minimum temperature goes above 22°C the fruit set is reduced. Under greenhouse conditions tomato crop can be grown for long duration of 10-12 months by cooling during summer months(April to July) and by heating the green houses during peak winter months (December to January) in Northern parts of the country (Harel *et al.* 2014).

3.2 Varieties

R 144, FA 574, FA 180, GS 600, Delphi, Astona, Cronos- Big fruited tomatoes of Indeterminate type.

Indeterminate varieties: Avtar, Indam hybrid, All Rounder, Arka Meghali, Arka Surabhi, etc. are suitable varieties.

HA818, NS Cherry 1, NS Cherry 2, T56 -cherry type.

3.3 Selection of varieties

- The most important pre-requisite for successful tomato cultivation in a greenhouse is selection of a suitable cultivar.
- The important characteristics related to cultivars include high yield, freedom from cracking, disease resistance, fruits high in number, resistant to green shoulder, higher shelf life and total soluble sugars.
- Indeterminate growing hybrids are suitable for green house cultivation.
- Hybrids growing at a height of 15 feet and above utilizing the vertical space in the GH and having a yield potential of 170 t/ha and more from a crop of 6months duration are best suited for GH cultivation (8-9 months).

3.4 Types of Cultivars

- **Beefsteak cultivation**: Large sized fruits with weight (180-250g), harvested individually with calyx. US cultivars like Trust and Match are high yielding and non-green back cultivars, susceptible to russeting. Popular varieties of this group are FA 574, FA 180 and FA 514.
- **Big fruited varieties:** Israel varieties which are suitable for the northern parts of our country. Average fruit weight id 120-160 g per fruit. Popular Israel varieties are R-144 (Daniela), FA-189, FA 179. Indian varieties -Naveen, Nun 7711, GC -600, Arka Vardhan and Arka Vishal. Fruits are harvested singly with calyx.

- **Cluster type/ Hand type:** They produce small fruits and harvested in clusters of 4-7 with the cluster stem still attached indicates the freshness of the fruits to the consumers. The clusters which are harvested when ready are marketed in a mesh bag, tray pack or in a single layer box and the range of maturity from breaker to ripe. Ex. HA 646, FA 556, FA-521
- **Cherry tomato:** Very small in size with a fruit weight of 12-20g with high TSS (6.8-7%). T-56 is available in India but the fruits are bigger in size and grown both open and in green house condition**.**

3.5 Season

1. Spring: Transplanting stage for spring in the greenhouse is last February or early march.
2. Autumn: The transplanting time to the green house is mid-august.

Fig. 3: Cherry tomato grown under protected cultivation

3.6 Nursery

Under protected structure nursery for green house tomato is raised, mostly, in the media which is soil less to produce disease free and virus free seedlings raised in plastic trays. The growing media is equal ratio between vermicompost and sand along with the farm yard manure and sterilized cocopeat. 98 cell seedling trays made of polypropylene material with drainage holes are used for seedling production. Trays are filled with the growing medium and tapped gently to fill the cells properly. Seedling

tray are drenched using copper oxychloride solution (3 g/l). Shallow depression of 5 mm depth are made in each cell. After treating the seeds with captan (0.2g/00g seeds) sowing is done and covered with a thin layer of the above growing medium and light irrigation is given. Filled trays are staked one above the other and covered with plastic sheet to create warmth and humidity required for germination. Seed germinate in 3-4 days after sowing. Shifted of the trays is done to the net house and it is placed on a raised bed, covered with a plastic sheet immediately after germination. Trays are watered twice a day to maintain appropriate moisture. Seedlings are drenched with 19:19:19 trace elements at 15 days after germination. Seedlings are sprayed using Acephate (0.75 g/L) to prevent infection by thrips. Seedlings are hardened by exposing them gradually to sunlight and by reducing the frequency of watering. Seedlings will be ready for planting after 25-30 days of sowing and 20g of seed is required for 1000 m^2 of greenhouse.

3.7 Growing bed preparation

Usually for tomato cultivation a raised bed is advisable. Well decomposed locally available organic manure at the rate of one bamboo basket per square meter of the bed is added, mixed thoroughly and the beds are levelled. In heavy soils-mixing sand up-to 25% is required to provide proper aeration in the root zone.

3.8 Disinfecting the growing beds

Disinfection is done with formaldehyde 4% and covered with a polyethylene (400 gauge) sheet and all ventilators are closed. The cover was removed after four days of formaldehyde treatment. To remove the trapped formaldehyde fumes completely, the raking is done to beds repeatedly before transplanting. The beds of following dimensions are preferable after fumigation. Top width of 90-100 cm and path width of 50 cm and height of 40cm.

3.9 Application of neem cake – trichoderma

Neem cake is made into powder, slightly watered to moisten and kept for two days and then mixed with Trichoderma formulation @1kg/100 kg neem cake. Mixture is closed with a plastic sheet and for every two days the mixture is turned once. Applied after 15 days of formaldehyde fumes are exhausted completely @ 200g/sq. m of bed and mixed thoroughly.

3.10 Fertilizer application

Commercial fertilizers supplying 50:50:50 NPK kg/ha applied to growing beds before fumigation.

3.11 Laying of drip line

The inline drip lateral must be arranged at the centre of bed. Inline drip lateral should have an emitting point for every 30 cm interval with discharge of 2 litre of water per hour. Uniform discharge of water to be noticed from the emitting point.

3.12 Mulching

Thick polyethylene mulch of 100micron (400-gauge thickness) film of 1.2m width is used to cover the planting bed. Make holes of 5cm diameter on the mulch film as pe the recommended spacing (60cm*45cm). The planting beds to be covered with the mulch film by checking the edges of the sheet and placing the soil inside.

3.13 Transplanting

Considering the temperature and light conditions, five to six-week-old seedlings were transplanted, during propagation. Size of ideal seedling is about 16cm and seedlings of 3000 are required for 1000sq.metre. Spacing adopted is 90*50*40cm.Water the beds to field capacity (moist but not drip wet) before transplanting. The seedlings should be sprayed with imidacloprid (0.3ml) solution one day before transplanting. Drench the seedlings with carbendazim (0.1%) solution on the day of planting in early morning hours. Transplant is preferred in the early morning or late evening. Seedlings may be planted in the holes made in the polythene mulch film so that the seedling does not touch the mulch film. The beds are irrigated on a daily basis until the establishment of the seedlings. Foggers are run if the humidity is lower. Beds are drenched with 0.3% copper-oxy chloride if mortality of the seedling are noticed.

3.14 Irrigation

Drip irrigation 10 days after transplanting. It is given daily to supply 2-3 l of water/ sq. m/day depending on the local weather conditions.

3.15 Fertigation

NPK is used in the ratio of 200:200:200 kg/ha through fertigation for entire crop growth period by using water soluble fertilizers. Supply WSF 19% each of NPK@ 3.75G/M for every fertigation weekly two times after transplanting from 3rd week. Sprayed with micro nutrient formulation (3g/l) 2 to 3 times once in 30 days starting from 60 days after transplanting. Crop is fertigated 2-3 times with calcium nitrate once in 15 days.

3.16 Growth regulators

Spraying planofix 4 times @2.5ml 10litre of water at n interval of 15 days from the initiation of first flowering is desirable to induce increased flowering and fruit set.

3.17 Application of fertilizers

The application of fertilizers is 250:250:250 kg NPK/ha in 10 equal splits at a 10days interval from the date of transplanting through the drip irrigation system.

3.18 Special intercultural operation

3.18.1 Staking: Staking which provide support with bamboo poles or wood sticks or any other sturdy material which keeps the foliage and fruits off the ground. It

can increase fruit yield, size, reducing the fruit rot,spraying can be done easily and harvesting**.**

3.18.2 Training: Each branch is trained along a separate plastic tube provided for the purpose. Branches are tied to the plastic tube with gunny twine so that the branches do not break due to weight of the foliage and fruits. Tying starts from 4 weeks after planting at weekly interval usually along with the pruning operation. There are 4 types of training: Vertical, Arch, V-shape, S-Hook system and Lateral.

3.18.3 Pruning: It starts from 20-30 days after transplanting at weekly interval. The main stem of tomato plant branches into two after the first flower cluster. Those two branches are retained and all other branches are removed. All the branches developing at the base of the stem also removed. Excess fruits removal will result in bigger size tomatoes fetching higher market price.

3.18.4 Cluster thinning: Tomato produces from one to about 19 flowers per cluster. Under good pollination conditions, about 6-8 of these flowers might from fruits. If higher number of fruits are allowed to set in a cluster, in terms of fruit size, shape, quality and uniformity will be declined. Therefore, thinning has to be carried out to maintain optimum number of fruits per cluster.

3.18.5 Suckering: Suckers will be developed between each compound leaf and stem in indeterminate types. These suckers are removed, leaving only the main stem as a growing point.

3.18.6 Topping: The small fruit that develops at the top of the plant at the growing point are removed before six weeks of the crop termination date to in order increase the size of the fruit which is already set in the lower part. Fruit requires 7-8 weeks from anthesis to harvest so small fruit will not have enough time to develop to maturity.

4. Pollination Requirement

4.1 Aided pollination: Due to the limited air movement and humidity which is high its needed for sure. The perfect pollinators are considered as Bumble bees, even under environmental stress condition.

4.2 Electric or battery-powered vibrators: There main work is to vibrate flowers cluster. During vibration sufficient pollen will be released which is necessary for pollination. This practice is done twice a day (10am to 11am and 2.0-3.0pm). Flowers are regularly shaken on daily basis. Air from a mist flower also has found effective.

4.3 Cooling and Heating of Greenhouses

4.4 Evaporative cooling system: The effective humidity in atmosphere is below 40%. Heating is provided from mid of December to January ending, to increase the temperature of the Greenhouse during nights, the temperature cannot fall below 14°C. Heating and cooling required for tomato crop is 10-12 months duration.

5. Harvesting

The first picking is done at 75-85 days after transplanting. Big tomato harvested singly with attached calyx. Summer months harvesting should be done in early morning or late evening to avoid postharvest losses. Cherry tomatoes are harvest with stem attached or some time singly with attached calyx.

6. Yield

The average yield of tomato under protected conditions may vary 6-8kg/m^2 if proper care is taken. The average yield per plant may be 3-4kg/plant with the fruit number of 40-60 / plant and fruit weight of 80-90g.

7. Findings from Earlier Research

Kundu (2014) conducted an experiment in protected cultivation of tomato of three different environments like glass house, poly house and net house with five different varieties namely Avinash, Naveen, Hybrid FA-574, R-144 and DT-7 of tomato and the results indicated that the cultivar Avinash recorded maximum plant length, number of leaves, number of branches, leaf area and number of fruits per plant. Mehta *et al.* (2020) has conducted an experiment on socio-economic impact in protected cultivation of tomato and reported that the crop productivity was increased by three times in comparison to the open-field conditions and this study revealed the positive impact of for Farmers by improving the socio-economic conditions. Tomato requires a relatively cool, dry climate for high yield and premium quality by Nicola *et al.* (2009). When the temperature falls below the 10°C, it causes problem with the pollen bursting, while the higher temperature causes premature fruit drops in tomato Singh *et al*. (2015). Lower temperature affects the crop production as there is problem with fertilization and leads to less fruit yield. The reported that high temperatures resulted in badly damaged or misshaped fruits making them unfit for market, while the red varieties tend to become more orange. These problems can be reduced by maintaining the temperature in protected cultivation. When the temperature rises above 30°C, both the pollen grain and stigma may dry out resulting in poor fruit set. Kumar *et al.* (2016) reveal that the cost of cultivation of tomato under poly houses was higher along with the net returns when compared to open field conditions. It has also been discussed about the yield which is accounting for 53.71% under protected than in the open field condition.

8. Conclusion

The main strategy of protected cultivation involves enhancing the socio-economic characteristics of the farmers and to generate a higher income, employment and a decent standard of living to the small and marginal framers of the state also to promote off-season cultivation. The recent studies revealed that protected cultivation has surely a great significant impact in enhancing the crop productivity and also

more in quality and quantity in 3 times than the open condition. It also helps in creating the employment generation to many people. The study also observed less income variability of polyhouse growers in comparison to the open-field growers and therefore, it can be concluded that the risk in polyhouse income was low, having a positive effect in the stability of income generated by the polyhouse growers in the study area. Therefore, it can be concluded that the protected cultivation has a significantly positive impact in improving the quality and in giving profitable income to the farmers.

References

[1] Harel, D., Fadida, H., Alik, S., Gantzand, S. and Shilo, K. 2014. The effect of mean daily temperature and relative humidity on pollen, fruit set and yield of tomato Grown in commercial protected cultivation. *Agronomy,* **4**(1): 167-177.

[2] Kundu, B. 2014. *Protected cultivation of tomato for assured livelihood.* Orissa University of Agriculture and Technology, Krishikosh.

[3] Kumar, P., Chauhan, R.S. and Grover, R.K. 2016. Economic analysis of tomato cultivation under poly house and open field conditions in Haryana, India. *Journal of Applied and Natural Sciences,* **8**(2): 846-848.

[4] Maitra, S., Shankar, T., Sairam, M. and Pine, S. 2020. Evaluation of gerbera (*Gerbera jamesonii* L.) cultivars for growth, yield and flower quality under protected cultivation. *Indian Journal of Natural Sciences,* **10**(60): 20271-20276.

[5] Mehta, K., Thakur, R.K. and Guleria, J.S. 2020. Socio-economic impact of protected cultivation on tomato growers of Himachal Pradesh. *Economic Affairs,* **65**(1): 01-07.

[6] Muthukumar, P. and Selvakumar, R. 2013. *Glaustas Horticulture*. 1st Edition, New Vishal Publications, New Delhi, pp. 544.

[7] Nicola, S., Tibaldi, G. and Fontana, E. 2009. Tomato Production Systems and their application to the Tropics. *Acta Horticulturae,* pp. 27-33.